

Pedagogic SEM User Manual

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4Y GP – Promo 60

2025 - 2026

Tutors: Almorice Jean, Cayez Simon, Houdellier Florent



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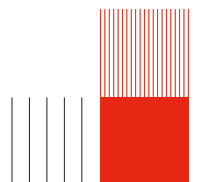
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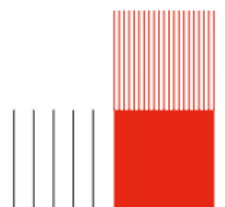
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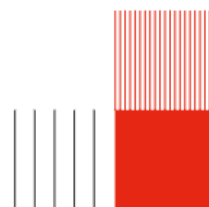
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Introduction

As part of our multidisciplinary project, we have finalized the creation of a fully educational Scanning Electron Microscope (SEM), allowing us to control each lens independently and making it suitable for teaching purposes. This SEM allows us to control each lens (condensers, objective, scanning lenses...), run a scan and retrieve an image from a secondary electron (SE) detector, view the electron beam inside the chamber in live via an optic camera, and finally run a simulation of the electrons trajectory inside the electronic column.

The educational SEM, which is originally a JEOL JSM 840, is located in the TP4 room of the physical engineering building. The program associated has been developed in Python 3.05 using the Visual Studio Code IDE and the PyQt6 library has been used to develop the user interface.

In this user manual, we show you how to use this SEM correctly. The first part is devoted to the hardware, including the necessary connections and physical actions. The second part is devoted to the software, including a presentation of the different features and how to use them.

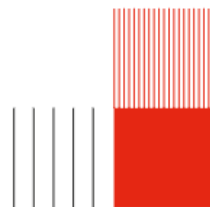
1 Equipment presentation

1.1 Overview

The SEM is composed of an electron gun to generate the beam, lenses used to focus this beam, a camera to see it in real time and an SE detector to acquire an image.

1.2 Emptying the column and the chamber

For the whole system to work, it is necessary to empty the column and the chamber. A Scroll primary pump and a turbomolecular secondary pump create a secondary vacuum in this system.



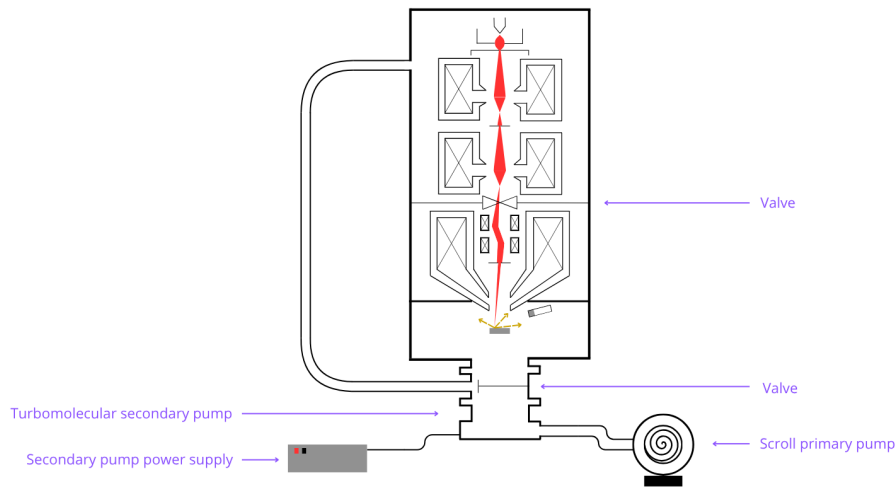


Figure 1: Schematic of the pump system to create a vacuum

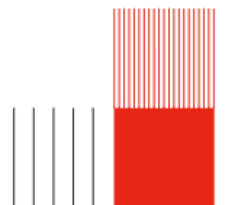
The step is usually done by the teachers.

1.3 Creating an electron beam

The electrons are emitted by a thermionic source. A tungsten filament is heated by a current of 1-2 A. The electrons are extracted from the valence shell of the tungsten atoms. They are then guided to the column by an conic piece called the Wehnelt.

In the following order, increasing slowly the potentiometer enables the user to have a current of around 1,9 A. The power supply is then turned on and its value is slowly increased for a final voltage value of 5 keV.

The box containing the whole supply system needs to be closed for safety. It can cause harm to the user and even death. It needs to be manipulated carefully.



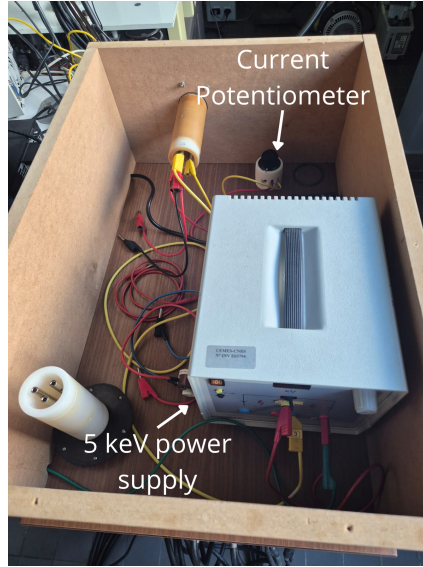


Figure 2: Image of the electron gun power supply

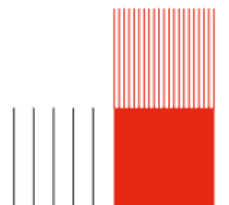
1.4 Lenses connections

The SEM is composed of five main lenses: 2 condenser lenses, 1 objective lens to focus the beam, 1 scanning lens in the X-direction and 1 scanning lens in the Y-direction to do a scan. Each lens is connected to one of the power supplies.

It is crucial not to exceed the following current and voltage values for each lens to not damage them.

Table 1: Lenses electrical data

	Condenser 1	Condenser 2	Objective	Scanning X	Scanning Y
Current (A)	1	1	5	0.500	0.500
Voltage (V)	10	10	10	0.700	0.700



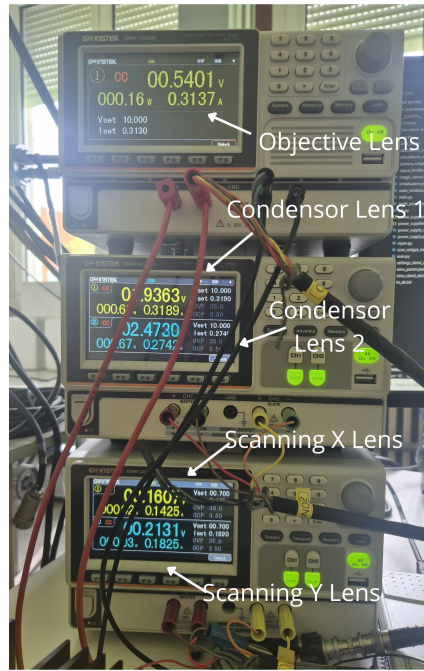


Figure 3: Images of the 5 power supplies that control the lenses system

All the values sent through the software can be checked in the power supplies.

1.5 SE detector usage

The SE detector is a box that **NEEDS TO BE CLOSED** when the system is on. It contains high-voltage power supplies that can cause damage to the user and even death.

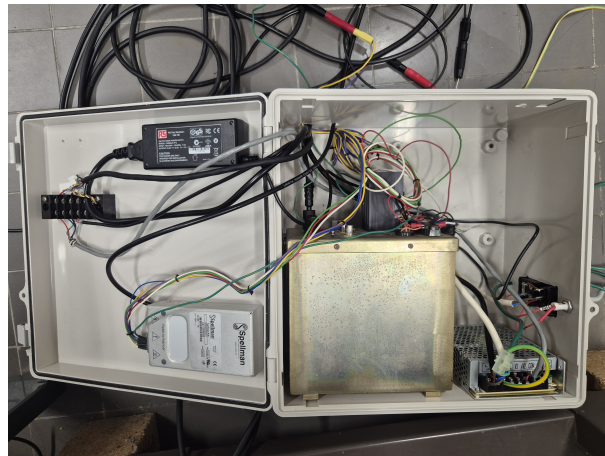
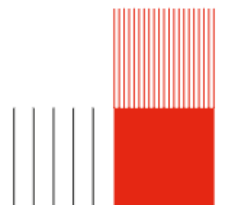


Figure 4: Image of the open SE detector

Moreover, the brightness can be changed with a little power supply that delivers 0 to 15V. **It is important not to set a voltage higher than 10 V.** The contrast can be adapted by a potentiometer. The voltage value associated can be read on the multimeter affiliated.

About the connections of the potentiometer: the purple wire of the SE detector needs to be connected to the green wire of the potentiometer, the blue wire to the yellow wire and the



ground wire to the white one.

About the connections of the brightness power supply : the blue wire needs to be connected to + terminal and the red wire to the - terminal.

The brightness adds an offset to the signal received by the scan. Its voltage ranges from 0 to 10 V. The contrast adds a gain to the system. The voltage ranges from 0 to 5 V. The potentiometer and the power supply can be seen below.

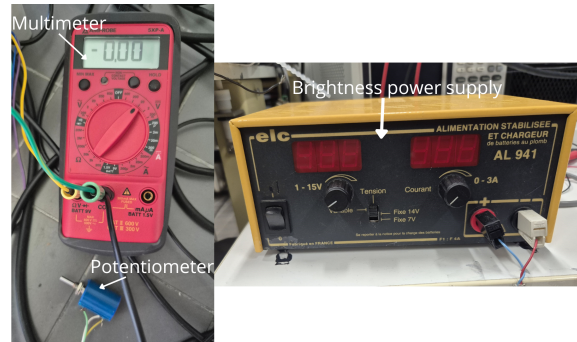


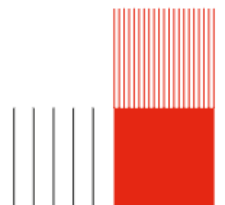
Figure 5: Image of the potentiometer controlling the contrast, and the power supply controlling the brightness

1.6 DAQ pins connections

The SE detector is connected to a USB-6210 NI DAQ card in non referenced single ended. The + terminal of the SE detector is connected to pin 17 (blue) and the - terminal is connected to pin 28 (red).



Figure 6: Image of the DAQ card



2 Program usage

2.1 Main Window

The software opens on the main window, which allows you to control which windows open for the 5 different features:

1. Multi Power Supply
2. Camera
3. Scan
4. Beam Simulation
5. Settings, which can be displayed by enabling the admin mode with the password “opc”

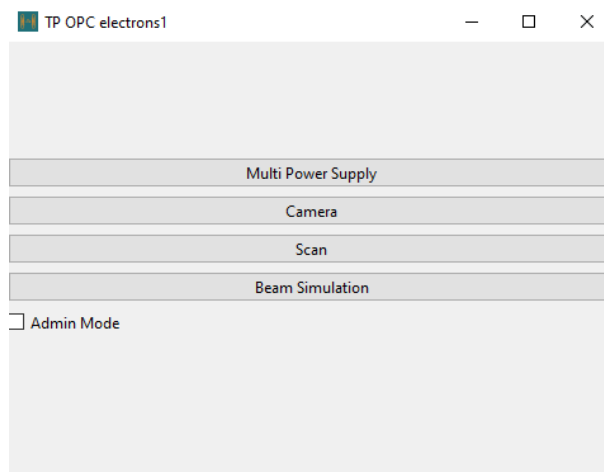


Figure 7: Main window

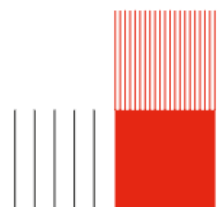
2.2 Multi Power Supply

This feature allows you to control the different configured lenses. Five lenses are already set by default (2 condensers, objective, scan X and scan Y), and each lens can be controlled as below.

At the top, the lens name, the power supply and the channel used are displayed.

In the center, the current set is controlled, either by the slider or the increment box. The voltage value is already set, but if needed voltage slider and increment box can be displayed by enabling admin mode with the password “opc” at the bottom of the window.

At the bottom, the set and measured values of current and voltage are displayed.



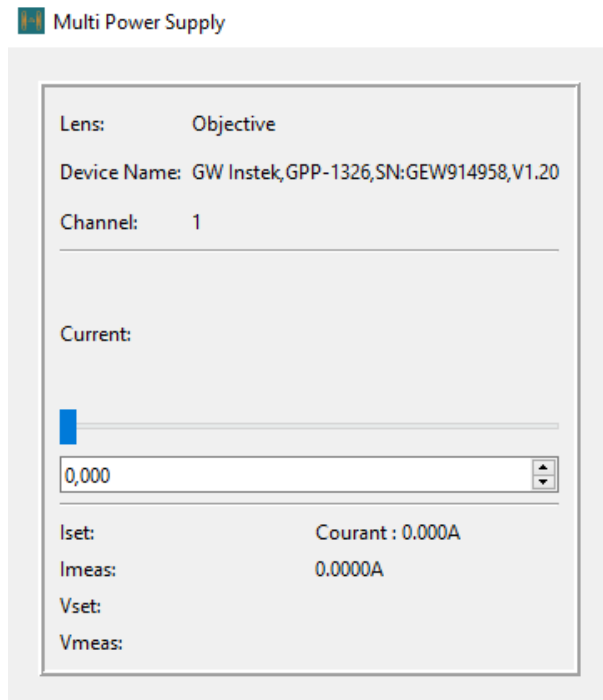


Figure 8: Power supply window

2.3 Camera

This feature allows you to view the image from the camera located on the side of the SEM. You can see the electron beam spot thanks to an electroluminescent plate.

You can move around the image using the arrow keys, zoom using + and - keys.

You can also capture the currently displayed image and save it.

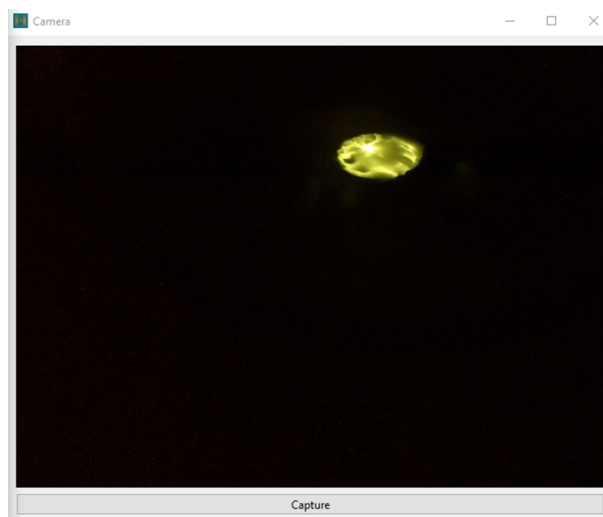
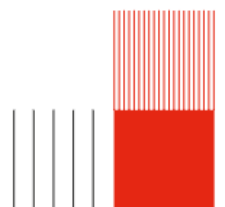


Figure 9: Camera window



2.4 Scan

This feature allows you to run a scan and display the image generated by the SE detector.

In the upper section, the image is displayed as the scan is performed.

In the lower section, you can configure the current range of the scan coils, the resolution of the image and averaging for each pixel. The SE voltage measured by the acquisition card (0-10 V) is also displayed, so you can adjust brightness and contrast accordingly. Three buttons allow you to start the scan, stop it while it is scanning if necessary, and save the image in PNG or TIFF format.

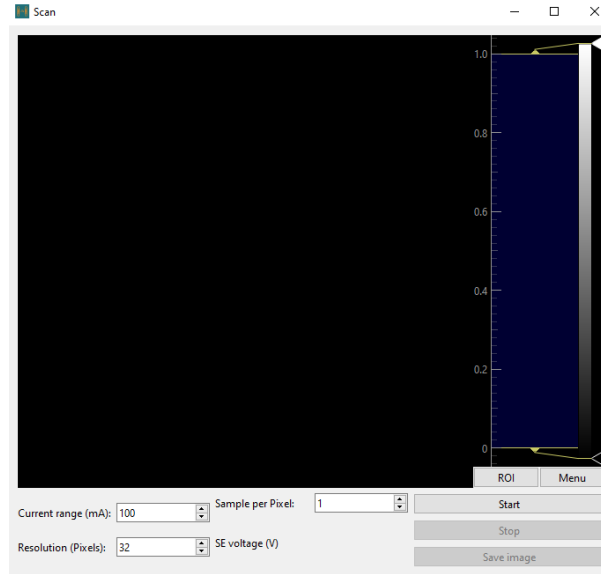


Figure 10: Scan window

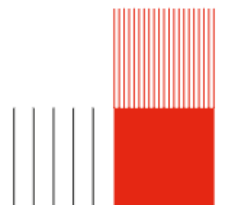
2.5 Beam Simulation

This features allows you to run a simulation of the paraxial electron ray trajectory inside the electron column.

In the upper section, the simulation is performed live in a graphic. You can choose either to display a single ray or the ray and its symmetrical counterpart for a more visual display by checking the box below the graphic. The passage through the different lenses is shown, and the simulation ray ends at the sample level.

In the lower section, the current measured within each lens and the maximum magnetic field associated are shown. Four boxes below allow you to change different parameters :

- The beam energy, which is normally set to 5000 eV (default value)
- The sample position, the default value (47 cm) is the plate position (distance between the electron gun and the plate)
- The initial conditions of the ray, which are the field and aperture. The default value (field 0 and aperture 1) is the marginal ray.



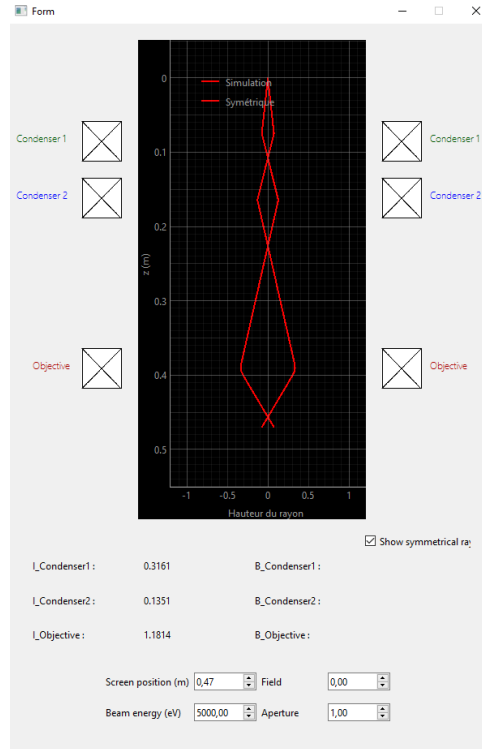


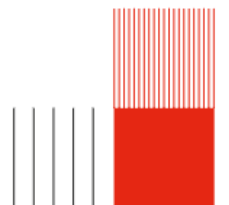
Figure 11: Beam simulation window

2.6 Settings

This is an optional feature which allows you to set manually the different lenses used and the power supplies associated. Indeed, five different lenses are already set by default in the multi power supply window.

At the top, you can choose the lens used, and the power supply associated. You can then enter the different values of minimum and maximum for current and voltage.

At the bottom, you can choose to add or delete the power supply to the list, check the list of entered power supplies, and once you are done, click on the "Finish" button. This will close the window.



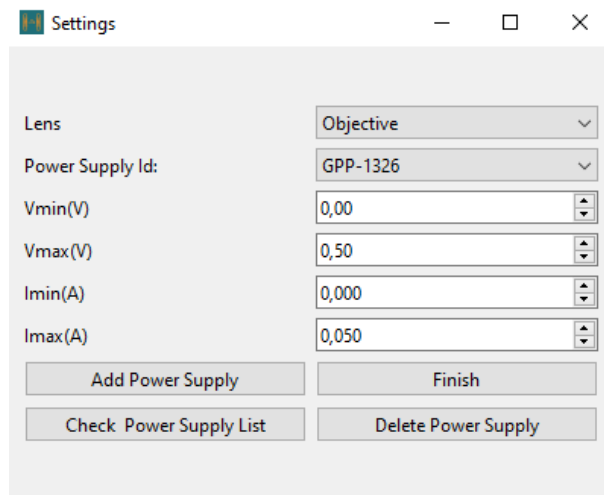


Figure 12: Settings window

Enjoy working on the SEM! :)

